

Renewable Energy Orchestrator — Architecture & Features

Oct 3, 2026 · @RIBHU Dubey

ET × Accenture AI Hackathon (Agentic Edition), Problem 4 — a dual-agent system that autonomously orchestrates solar, wind, battery and grid-market decisions every 15 minutes, with a built-in evaluator that catches unreliable decisions automatically.

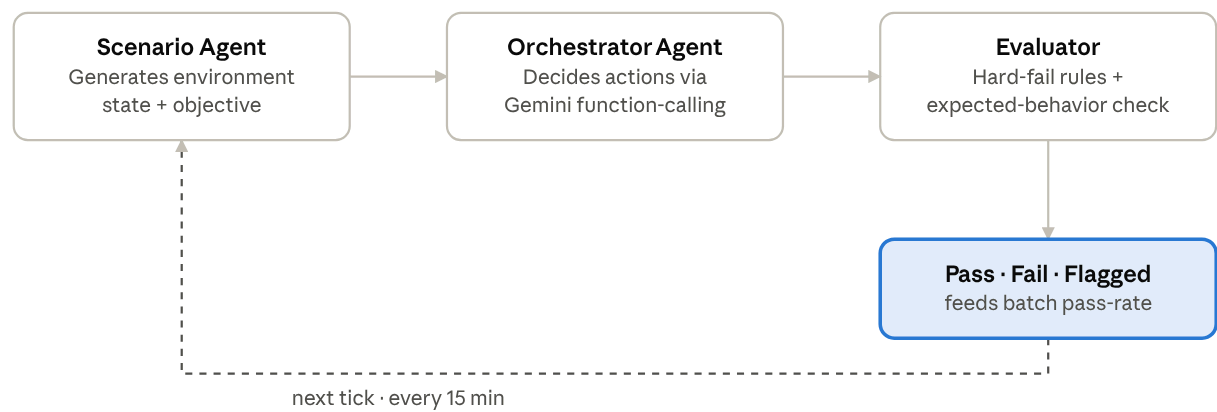
Problem & Approach

A utility operates 5 solar farms, 3 wind farms, 2 battery systems, industrial consumers, and grid/market access, with conditions shifting every 15 minutes — clouds cutting solar, prices spiking, a battery going offline, demand surging. The team chose this problem (Utilities, Problem 4) over Banking Regulations, the Prompt Injection Firewall, and the Retail Promotion Planner because it has the strongest Agentic Capability story: a genuine continuous decision loop under shifting state, not a one-shot classification or recommendation task.

The key risk is a self-verification gap: without energy-domain expertise, a decision can't be judged correct by instinct the way a retail discount could. The architecture below closes that gap by separating decision-making from verification, and grounding verification in publicly documented grid-operating principles rather than invented rules.

System Architecture

Three agents turn raw grid data into a checked decision every tick
Runs once per 15-minute tick



three agents, one tick, one checked decision

The Scenario Agent's expected-behavior tag is generated alongside the environment state but withheld from the Orchestrator — only the Evaluator sees it, which is what makes a decision checkable without inventing energy expertise.

Key Actions & Decisions

Each tick, the Orchestrator Agent decides: battery actions (charge, discharge, reserve, or hold, per battery, in MW), a market action (buy, sell, hold, or delay sale, in MW), how much solar/wind to curtail (if any), and whether to trigger demand response.

Declared objective	What "optimal" means
cost_efficiency	Cheapest available source dispatched first (merit order)
min_carbon	Lowest-emission source used, even if not cheapest
max_reliability	Demand met first, at any available cost
max_profit	Sell surplus when prices are high; avoid buying unless needed

Declared objective	What "optimal" means
None declared	Default Priority fallback — reliability first, cost-efficiency second: fixed, not learned, defensible because reliability is treated as non-negotiable in real grid operations

Declaring the objective per scenario, rather than fixing one global objective, is what demonstrates the hackathon's F2 requirement: determining the optimal cluster of options after establishing optimality criteria for a specific situation.

Model Usage

The Orchestrator Agent calls Gemini (`gemini-3.5-flash-lite`) with forced function-calling (`tool_config mode ANY`), so every response is a structured `submit_decision` call, never free text that would need separate parsing.

The system prompt grounds every decision in four real grid-operating principles rather than letting the model invent strategy: merit order dispatch (cheapest source first), battery charge/discharge logic (charge when cheap or in surplus, discharge when expensive or scarce), curtailment as a last resort, and reliability as a hard constraint that overrides cost or profit. The model also writes a step-by-step `reasoning` field explaining which principle(s) drove each action — used by both the Evaluator and the live demo narrative.

Transient Gemini free-tier `503` errors ("model experiencing high demand") are retried automatically up to 5 times with exponential backoff before surfacing to the user.

Key Features Implemented

Mapped against the capabilities and inputs the problem statement names:

Capability	Status
Battery actions — charge, discharge, reserve	Implemented
Market actions — buy, sell, delay	Implemented

Capability	Status
Renewable actions — curtail wind/solar, prioritize clean generation	Implemented (curtailment direct; "prioritize cleaner generation" via the min_carbon objective)
Demand management — trigger demand response, reduce/shift loads	Partial — demand-response trigger implemented; load shifting not modeled
Maintenance actions — delay/schedule/dispatch	Not implemented
Multi-objective handling (cost, carbon, reliability, profit)	Implemented — per-scenario declared objective + Default Priority fallback
Renewable generation inputs (solar/wind output + forecast)	Implemented
Grid inputs (demand, frequency, transmission constraints)	Implemented
Battery inputs (state of charge, efficiency, rates, degradation)	Implemented
Market inputs (prices, carbon price, demand-response incentive)	Implemented
External inputs (weather, storm alerts, maintenance schedules)	Partial — weather/storm modeled; maintenance schedules not yet an input

9-Blocker Self-Assessment

Currently claimed position: D2 × F2, backed by the Evaluator's pass/fail/flagged output across batch runs spanning all 4 difficulty profiles and all 4 objectives.

	F1 — single action cluster	F2 — objective-aware, uncertainty-adjusted	F3 — multi-tick simulation
D1 — structured input, acceptable output	Demonstrated	Demonstrated	Not built

	F1 — single action cluster	F2 — objective-aware, uncertainty-adjusted	F3 — multi-tick simulation
D2 — structured input, high reliability	Demonstrated (batch pass-rate)	Demonstrated (batch run, mixed objectives)	Not built
D3 — multimodal input, high reliability	Not built	Not built	Not built

D3 (multimodal input) and F3 (multi-tick simulation) are open next steps, not current claims — see the Next Steps document. The hackathon penalizes over-estimation equally with under-estimation, so this position should not move until the demo evidence backs it up.

Responsible AI & Reliability

Every decision is checked against six hard-fail rules, regardless of declared objective, plus an objective-specific expected-behavior check the Scenario Agent attaches but never shows the Orchestrator:

Rule	What it catches
1	Curtailing renewables while also buying from the grid in the same tick
2	A battery discharged below its minimum safe charge
3	Demand exceeding available supply (hard reliability violation)
4	Selling to the grid while industrial demand is unmet
5	Charging the battery at a daily price peak
6	Curtailment used when it isn't actually a last resort

A decision that fails a hard-fail rule is marked fail; one that only diverges from the expected objective-specific behavior is flagged rather than failed. This is what lets the team demonstrate — not just claim — Scalability, Responsible AI & Robustness and Agentic / Autonomous Capability with actual evidence rather than assertion.